Service Operations Management Analysis of a Third-Party Logistics Provider in Saudi Arabia

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1.0 INTRODUCTION TO THE STUDY

The logistics sector in Saudi Arabia has experienced significant growth in recent years, driven by the country's strategic vision to become a global logistics hub under Vision 2030. Efficient logistics and supply chain operations are essential to support trade, enhance customer satisfaction, and improve overall economic competitiveness. Service operations management plays a critical role in ensuring that logistics firms deliver reliable, timely, and high-quality services to their customers.

This study focuses on a leading third-party logistics (3PL) provider operating in Saudi Arabia. The company offers a wide range of services, including warehousing, transportation, inventory management, packaging, and distribution. These services are essential for businesses that rely on efficient supply chain operations to deliver products to customers across different regions. The company operates through a network of warehouses and distribution centers, primarily located in major cities such as Jeddah and Riyadh, enabling it to serve both local and international markets.

The logistics industry in Saudi Arabia is highly competitive, with major players such as Aramex, DHL, and SMSA Express competing to provide efficient and reliable services. These companies differentiate themselves through service quality, delivery speed, technological integration, and operational efficiency. The selected company positions itself as a service-oriented logistics provider that emphasizes quality, reliability, and continuous process improvement.

Despite its strong operational presence, the company faces challenges related to service quality, particularly product damage during handling and transportation processes. Such issues directly affect customer satisfaction, increase operational costs, and impact the company's reputation. Addressing these challenges requires a structured approach to service operations management, focusing on process improvement, quality control, and operational efficiency.

This project aims to analyze the service operations management practices of the selected logistics company, with a particular focus on quality management, process design, and operational performance. By examining key areas such as strategy, capacity planning, service design, and quality management, the study provides insights into how the company can improve its service delivery and achieve sustainable competitive advantage in the logistics industry.

2.0 STRATEGY OF THE COMPANY

The strategy of a service organization determines how it competes in the market and delivers value to its customers. In the logistics industry, firms typically compete based on cost efficiency, service quality, speed, reliability, and flexibility. The selected company, a leading third-party logistics (3PL) provider in Saudi Arabia, has developed a strategy centered on operational excellence and high-quality service delivery.

2.1 Mission of the Company

The company's mission is to become a leading strategic logistics service provider in the region by delivering reliable, timely, and value-added logistics services while exceeding customer expectations through accurate inventory control, complete product care, and efficient operations.

2.2 Core Competence

The company's core competence lies in its ability to provide integrated end-to-end logistics solutions, including warehousing, transportation, customs clearance, and distribution under one system. It has strong capabilities in managing complex supply chains, particularly in sectors such as healthcare, FMCG, and retail, where product handling and quality are critical.

2.3 Types of Strategies

Organizations typically follow one of three main strategies:

- **Low-cost strategy** – focusing on cost reduction
- **Differentiation strategy** – focusing on superior service quality
- **Response strategy** – focusing on speed and flexibility

2.4 SWOT Analysis



Figure 1: SWOT Analysis

2.5 Selected Strategy of the Company

The company primarily follows a differentiation strategy, as it focuses on delivering high-quality logistics services, reliability, and operational excellence. This is supported by its emphasis on complete product care, accurate inventory management, and value-added services rather than competing solely on price.

At the same time, the company also incorporates elements of a response strategy, as it aims to provide fast delivery and flexible logistics solutions across the Kingdom through its distributed warehouse network.

2.6 Functional Strategy and Alignment

An example of a functional strategy is the company's **operations and quality improvement strategy**, which focuses on reducing product damage and improving service reliability. The use of Lean Six Sigma methodologies (DMAIC) and FMEA tools in this study reflects the company's commitment to continuous improvement.

This functional strategy aligns with the overall differentiation strategy by ensuring high service quality, minimizing errors, and enhancing customer satisfaction.

2.7 Productivity Measurement

The company measures productivity using key performance indicators (KPIs), including:

- Damage rate of shipments
- Lead time performance
- Order accuracy
- Quality compliance rates

These metrics are supported by advanced warehouse systems that provide real-time inventory tracking and performance monitoring. However, measuring productivity in logistics can be challenging due to variability in demand, human involvement, and service-based operations. Therefore, the company relies on data-driven tools and performance dashboards to improve measurement accuracy and decision-making.

3.0 CAPACITY PLANNING

3.1 Capacity Management Framework

The company has a tech-based capacity management strategy, which aims to flexibly scale operations based on the changing demand. Elasticity is the key to this framework, allowing it to expand or contract through automation and infrastructure instead of fixed labour up increases. This strategy is based on **three pillars:**

1. Automation (Hardware): Advanced robotics systems like AutoStore in new logistics parks improves the storage density and throughput and reduces the number of manual operations. Its modular design would allow it to utilize space and scale storage capacity flexibly and in response to the variability in demand.
2. AI-Based Forecasting (Software): Decision-support systems such as Galys Monitor will give real-time access to inventory, productivity, and workflow KPIs. It helps to plan demand and optimise workflow, which enables more reliable capacity planning and responsiveness in operations.
3. Strategic Real Estate Deployment: A distributed warehouses network in major Saudi cities is used to provide geographic coverage and redundancy, with the central Riyadh Logistics Park being used as a high capacity operational centre to consolidate inventory and order fulfilment services.

3.2 Supporting Operational Practices

Capacity management is strengthened by operational practices that monitor workflow and inventory in real-time, and cross-docking, as well as increase the speed of throughput and minimize idle storage. These

measures, together with automated systems, contribute to the high operational efficiency even in the peak demand periods.

3.4 Current Capacity Status

- ***AutoStore robotics integration*** in the Riyadh plant has enhanced storage efficiency and throughput, raising system availability rates above 99% and productivity up to 4 times higher than conventional systems.
- ***Scalable distribution*** is supported by a network of various warehouses throughout the country in seven major cities and allows the delivery coverage of one to two days within the Kingdom.

These investments imply sufficient capacity to support the current level of demand of most typical logistics operations. On special peak times like national health campaigns or seasonal surges, additional solutions may be required like temporary workforce increases or outsourcing to maintain consistent service levels.

3.5 Short-term vs Long-term Capacity Planning

- ***Short-term*** planning includes operational changes that can be quickly introduced into the existing facilities. These include temporary labour, alternating shift, adjusting delivery schedules, and rearranging the transportation plans to flatten the capacity utilisation.
- ***Long-term*** planning focuses on strategic and structural investments, like the construction of the Riyadh Logistics Park and the expansion of warehouses across the country. Long term planning establishes the base of structural capacity and short term planning manages fluctuations around the capacity base.

3.6 Capacity Strategy

The business strategy fits a **lead capacity strategy**, whereby capacity is actively planned before the anticipated demand through automation, scalable real estate and digital tools. This is a proactive stance where the company is able to remain **agile** and prevent bottlenecks before they become a reality instead of responding to the demands once they elevate.

3.7 Fit to Competitive Strategy

Capacity management directly supports the company's overall responsiveness and differentiation on the 3PL market. Automated systems and predictive software investments increase delivery speed, reliability during demand spikes, and precision in order fulfilment, which are vital competitive advantages in both FMCG and healthcare logistics. High capacity readiness supports quick turnaround time and minimizes stock outs, particularly of essential healthcare products, which builds the reputation of the company in regard to reliable logistics services.

4.0 PROCESS SELECTION AND FACILITY LAYOUT

4.1 Process Flow Diagram

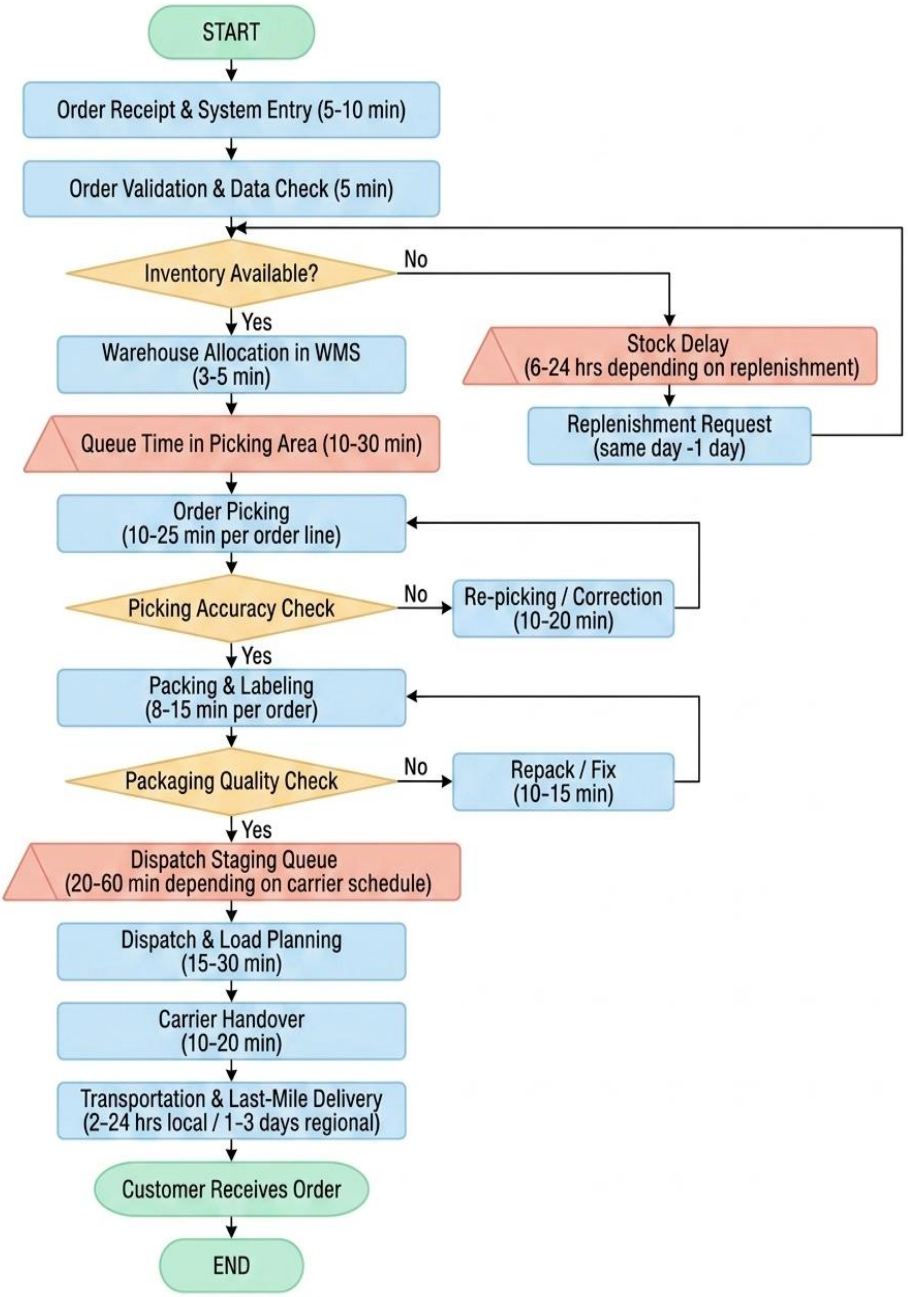


Figure 2: Process Flow Diagram of Logistics

4.2 Process Design

The operation is designed in a **repetitive process**, which is applicable to **high-volume** and **standardized** order fulfillment in logistics of pharmaceutical and FMCG. It manages massive order quantities and thousands of SKUs, and it needs to have organized and uninterrupted working processes. The **variety of products is high**, which includes FMCG goods and temperature-sensitive pharmaceuticals, which demand special handling conditions. Although there is this **diversity**, standardization of processes has been done to provide efficiency and accuracy.

Modular systems, ***zone-based picking*** and ***scalable automation*** provide **flexibility** to support peak demand periods (e.g. seasonal spikes) in the facility. One such case is the use of automated goods-to-person systems (AutoStore-type solutions) which enhance the speed of picking and minimizes manual movement.

Advantages:

- Good efficiency and productivity.
- Rapid and quick response to orders.
- Excellent precision and repeatability.
- Reduced costs of operation due to automation.

4.3 Facility Layout

The plant adopts a **hybrid design** with **high-density storage systems,** such as **push-back racking,** to optimise vertical space utilisation and maintain rapid access to SKUs, especially during **peak times.** **Distinct zones** are dedicated for order preparation, pharmaceutical storage, and fast-moving consumer goods, enabling streamlined workflows and minimising handling times.

The layout aligns with the company's objectives of secure and accurate delivery through automation and regulated areas, while maximizing space with dense vertical storage and narrow-aisle systems. It is flexible by cross-docking and modular automation, safe by controlled access and computerized tracking, more efficient and ergonomic by structured workflow, and environmentally focused by waste reduction and congestion mitigation with energy-efficient systems and logistic planning.

4.4 Layout Evaluation and Strategy Alignment

The firm follows a differentiation and responsiveness strategy, supported by a layout that reduces handling through automation, minimises travel time between zones, and ensures controlled storage to prevent product damage. The facility design enables a smooth workflow from receiving to dispatch, providing timely, accurate, and dependable distribution. Overall, the automated hybrid layout enhances speed, scalability, and service quality, particularly for pharmaceutical and FMCG supply chains.

5.0 SERVICE BLUEPRINT

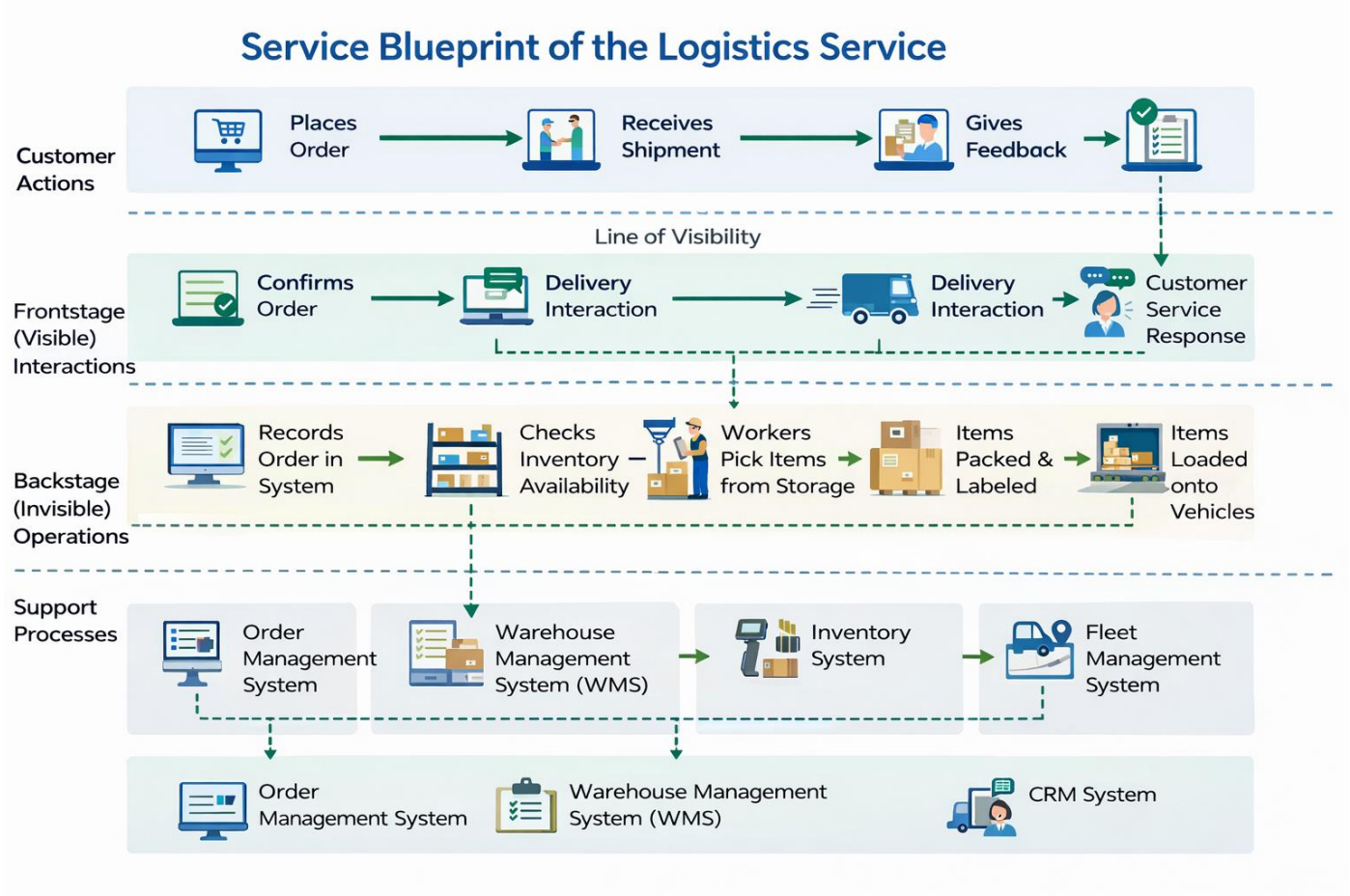


Figure 3: Service Blueprint

The service blueprint illustrates the complete logistics service process, starting from order placement to final delivery and post-service feedback. Activities above the line of visibility represent interactions that are directly experienced by the customer, such as order confirmation and delivery. In contrast, activities below the line of visibility include internal operations such as picking, packing, and loading, which are essential for service delivery but are not visible to the customer.

The blueprint highlights the importance of coordination between frontstage and backstage activities to ensure service quality and efficiency. It also demonstrates how support systems, such as warehouse management systems and tracking technologies, enable smooth service operations. Any inefficiencies in backstage processes, such as improper packing or handling, can directly affect customer satisfaction, as seen in the case of product damage.

6.0 SERVICE DESIGN

Service design for the chosen 3PL firm is designed to offer integrated and efficient services. It offers consistent logistics services to various industries, especially food, beverages, confectionery, and healthcare sectors. This organization employs both standardized and flexible service designs to meet its objectives of efficiency and effectiveness.

This organization has an end-to-end integrated service design strategy where logistics services are delivered as one process that covers logistics services from processing orders through warehousing, managing inventory, transporting, and delivering. Another aspect of its service design is the modular strategy where services are offered in various modules to clients depending on their needs.

A significant aspect of service design is that the operations are highly technology-oriented, with the deployment of WMSs, real-time tracking systems, and auto-storages like AutoStore. Technology adoption results in better inventory management, increased visibility, and minimal human errors. Moreover, the company's service design involves standardization of operations, especially in picking, packing, and shipping. On the other hand, flexibility has been achieved through scalability in operations, decentralized warehouses, and adaptive deliveries.

The company is successfully executing its service design based on the differentiation strategy that has been adopted. Product care, accuracy, and high-quality service are some aspects of differentiation, whereas fast deliveries, adaptable logistics solutions, and real-time tracking systems are aspects of response. For instance, the availability of temperature control zones for pharmaceutical goods guarantees compliance and safe storage of items, thus improving the quality of services offered.

When introducing a new service offering, there are several issues that the organization should be mindful of regarding its compliance with laws, ethics, and sustainability. Regarding the laws, the organization is required by the laws of Saudi Arabia to ensure compliance with the transportation, customs clearance, labor, and pharmaceuticals acts. Ethical issues involve ensuring a safe environment for the employees, minimizing damages on products, being transparent while delivering the services, and customer data protection.

In fact, the firm takes into account all of these factors while designing its services. The first aspect to consider is the legal one, as the company guarantees that its products will be handled within a regulated manner, especially in the case of pharmaceuticals and temperature-sensitive products. Ethical factors are related to minimizing the damage caused by improper handling, improving packaging, and enhancing quality controls, thus contributing to increased customer confidence in the firm. Moreover, from a

sustainability point of view, the organization introduces route optimization aimed at lowering fuel consumption, utilizes efficient warehouse systems, and avoids unnecessary rework and waste by optimizing operations. Automation is another important factor to consider, as it significantly reduces handling costs and eliminates any errors related to the manual treatment of products.

7.0 LOCATION

The company operates through a strategically distributed network of logistics facilities across Saudi Arabia, with major operations located in Riyadh, Jeddah, and Dammam. The Riyadh Logistics Park serves as the central hub, where high-volume inventory is stored and managed, while regional warehouses support distribution and last-mile delivery.

7.1 Evaluation of the Location

The company's location strategy is highly effective and well-aligned with its operational goals. By positioning its facilities in major economic cities, the firm is able to reduce delivery lead times and improve service responsiveness. For example, the central location in Riyadh allows efficient nationwide distribution, enabling delivery within 1-2 days across most regions.

In addition, Jeddah plays a critical role in connecting the company to international supply chains through its seaport, while Dammam strengthens access to the Eastern region. This combination creates a balanced logistics network that supports both domestic and international operations.

7.2 Factors Influencing Location Decisions

The company's location decisions are influenced by several key factors:

- **Proximity to customers:** Distributed warehouses across Saudi Arabia reduce delivery times and enable 1–2 day nationwide coverage.
- **Access to transportation infrastructure:** Strategic locations near highways and logistics corridors support efficient movement of goods across the Kingdom and integration with ports and airports.
- **Labor availability:** Urban areas provide access to a skilled workforce
- **Cost considerations:** Balancing facility costs with transportation efficiency
- **Government initiatives:** Vision 2030 supports logistics infrastructure and investment
- **Technology capability:** Modern logistics parks support automation and advanced systems

These factors reflect standard location decision models in operations management, where both cost and service level considerations are balanced.

7.3 Alignment with Strategy

The company follows a differentiation strategy with elements of responsiveness, focusing on high service quality and fast delivery. Its location strategy strongly supports this by enabling quick response times, improving service reliability, and ensuring flexibility through a decentralized network.

Overall, the company's location decisions contribute directly to its competitive advantage by improving operational efficiency, enhancing customer satisfaction, and supporting long-term growth in the logistics sector.

8.0 QUALITY MANAGEMENT

Quality management is a critical success factor for the company, as it directly impacts customer satisfaction, operational efficiency, and competitive positioning. Given the company's differentiation strategy, maintaining high service quality is essential, especially in addressing issues such as product damage during handling and transportation.

8.1 Dimensions of Quality

In logistics services, quality can be evaluated through multiple dimensions. The company focuses on the following:

- **Reliability:** Consistent and error-free order fulfillment supported by structured warehousing systems and automation.
- **Responsiveness:** Fast order processing and distribution supported by nationwide warehouse networks enabling 1–2 day delivery across Saudi Arabia.
- **Assurance:** Safe handling of goods, particularly pharmaceuticals and FMCG products through controlled processes and high-density storage systems.
- **Tangibles:** Quality of packaging, facilities, equipment, and automated storage systems such as push-back racks and pallet shuttle solutions that improve efficiency and space utilisation.
- **Customer focus:** Ability to respond effectively to customer needs supported by real-time visibility systems and logistics optimisation tools.

These dimensions align with the SERVQUAL model and reflect the importance of both operational performance and customer experience in service organizations.

8.2 Quality Improvement Practices

The company has implemented several practices to improve quality:

• **Lean Six Sigma (DMAIC):** Used to identify root causes of defects and improve processes
• **Failure Mode and Effects Analysis (FMEA):** Helps in identifying risks and preventing failures
• **Automation systems:** Reduce human error and improve consistency
• **Real-time tracking technologies:** Enhance visibility and control
• **Employee training programs:** Improve handling and operational accuracy

Despite these efforts, the issue of product damage suggests that further improvements are needed. The company can enhance quality by:

- Improving packaging design and materials
- Increasing employee training focused on quality awareness
- Implementing stricter handling procedures
- Using predictive analytics to identify potential risks

These improvements would shift the company from reactive quality control to proactive quality management.

8.3 Cost of Quality

The company incurs four main categories of quality-related costs:

- **Prevention costs:** Training, system upgrades, and process improvements
- **Appraisal costs:** Inspection and quality checks
- **Internal failure costs:** Rework, repackaging, and handling damaged goods before delivery
- **External failure costs:** Returns, complaints, compensation, and reputational damage

Among these, external failure costs are the most critical, as they directly affect customer trust and long-term business performance.

8.4 Quality Tools Used

The company uses several quality management tools, including:

- FMEA for risk assessment
- Key Performance Indicators (KPIs) such as damage rate and order accuracy

- Process analysis tools to improve workflow
- Cause-and-effect (fishbone) diagrams to identify root causes

These tools help the company move towards continuous improvement and better decision-making.

8.5 Inspection and Benchmarking

Inspection plays an important role in ensuring quality standards are met. The company applies inspection at multiple stages, including:

- Receiving goods
- During picking and packing
- Before shipment

This is especially important for high-risk products such as pharmaceuticals.

Benchmarking is also used by comparing performance with industry leaders such as DHL and Aramex. This allows the company to adopt best practices, improve service quality, and remain competitive.

8.6 Role of Inspection in Operations

Inspection is applied at key control points throughout the logistics process. However, relying too heavily on inspection can increase costs and reduce efficiency. Therefore, the company should focus more on building quality into the process through better system design, automation, and employee training.

This approach ensures that errors are prevented rather than detected after they occur, which is more effective in the long term.

9.0 CONCLUSIONS

Upon careful consideration of the results of the service operations analysis, it can be observed that the firm has managed to create a robust structure backed by innovative technology, standardization of processes, and strategy alignment. Having entered the highly competitive logistics market, the company implements the strategy of differentiation by offering premium-quality service, reliable performance, and integrated logistics solutions, but at the same time, it includes features of responsiveness by means of providing fast and flexible services.

It can be concluded that the company is implementing the selected strategy rather successfully in most cases. In particular, the employment of such technologies as automated processes, real-time tracking, and decentralized warehousing facilities contributes to both the differentiation and responsiveness of the company. Moreover, structured process management and quality control processes ensure effective performance. Nevertheless, as previously mentioned, the problem of damaged products may prove to be an impediment to full strategy implementation.

As far as the company's future development is concerned, the firm is projected to experience tremendous growth owing to the increasing demand for logistics solutions in Saudi Arabia within the framework of the country's Vision 2030 and the growing popularity of e-commerce. In turn, the firm will probably build a bigger warehouse network and implement some innovative technologies. In the event that these aspects are managed successfully, the company will be able to establish itself as one of the leaders of the 3PL market.

Several suggestions regarding the firm's future performance could be put forth. First of all, there is a need to pay particular attention to improving the quality control system, namely to reducing damages when handling products and delivering them. This may involve conducting more rigorous inspections, improving packaging methods and training employees properly. Secondly, additional investments should be made in technologies, especially AI for demand forecasts and predictions.

Furthermore, the company needs to concentrate more on sustainability measures such as using fuel-efficient or even electric delivery trucks, reducing the amount of waste generated through packaging, and establishing carbon measuring systems. Such steps would ensure that not only is there reduced environmental damage but also that the business stays consistent with international standards and regulations. Another way in which the company's performance could be improved would be through the expansion of performance measurement metrics.

To summarize, the company shows a great degree of strategic alignment with regards to service operation processes, but continuous improvement is crucial in order for it to remain competitive. Through improving operational processes by tackling product damage and focusing on innovation and quality assurance, among other measures, the company can ensure that it grows in the long run.

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